

Incidence and prognosis of cardiac conduction system diseases in hypertension: the STEP trial

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Patients with cardiac conduction system diseases (CSD) may have increased incidence and mortality of cardiovascular events. Here we report a post hoc analysis of the Strategy of Blood Pressure Intervention in the Elderly Hypertensive Patients (STEP) randomized clinical trial (ClinicalTrials.gov number, NCT03015311) concerning the effect of intensive blood pressure (BP) control on the incidence of new-onset CSD and the prognostic implications of preexisting or new-onset CSD. The incidence of new-onset CSD was similar in the intensive ($n = 205$, 6.42%) and standard ($n = 188$, 5.94%) treatment arms. Participants with preexisting CSD had a higher risk for acute decompensated heart failure. Increased age, male sex and increased body mass index were independently associated with increased risk for new-onset CSD. Our results suggest that intensive BP control may not reduce the incidence of new-onset CSD compared with standard BP control.

Conduction system diseases (CSD) are commonly observed on the surface of 12-lead electrocardiogram (ECG), especially among older adults^{1,2}. Numerous studies have evaluated the prognosis of CSD on the basis of ECG^{1,3–5}. Progression of CSD can lead to pacemaker implantation, heart failure, cardiovascular events, and even death^{1,3–5}. For instance, previous studies have linked left and right bundle branch block (LBBB and RBBB, respectively) to elevated cardiovascular morbidity and mortality^{6–9}. Several studies have identified that left anterior branch block^{3,10}, left posterior branch block^{3,11}, nonspecific intraventricular conduction delay^{3,12} and first-degree atrioventricular block^{13,14} all indicate unfavorable outcomes. In addition, the only existing therapy is pacemaker implantation, for which there is no established prevention strategy¹⁵. Understanding the factors that prevent or delay CSD can reduce the likelihood of pacemaker implantation for some patients at high risk.

A previous histological analysis of autopsy specimens identified fibrosis as a driving force behind cardiac conduction system abnormalities¹⁶. Moreover, there are known mechanistic links among hypertension, inflammation, fibrosis and abnormal conduction¹⁷.

In several studies, hypertension was found to be strongly associated with CSD^{18–21}. While results from the Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial and another study indicate reductions in incident CSD and intraventricular conduction delays with lisinopril and losartan therapy, respectively^{22,23}, there is limited data on the impact of intensive blood pressure (BP) control on the incidence of CSD in patients with hypertension. A post hoc analysis of the Systolic Blood Pressure Intervention Trial (SPRINT) showed that intensive BP control was associated with decreased risk of new-onset left ventricular conduction disease (LVCD)²⁴. However, this study was confined to adults in the United States, and did not include adults with diabetes mellitus. Whether intensive BP control reduces the new-onset of CSD in older Chinese with hypertension is unclear. In this Letter, therefore, we aimed to test these hypotheses on the basis of the STEP data. Additionally, we also investigated the prognostic implications of preexisting or new-onset CSD, and the influence of various clinical characteristics on the occurrence and progression of CSD in older Chinese patients with hypertension.

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Table 1 | Baseline characteristics of the study participants stratified by the status of CSD

Characteristics	Free of CSD; <i>n</i> =5,966	New-onset CSD; <i>n</i> =393	Preexisting CSD; <i>n</i> =766	<i>P</i> value (free of CSD versus preexisting CSD)	<i>P</i> value (free of CSD versus new-onset CSD)
Age, years	65 (62–69)	66 (63–70)	67 (63–71)	<0.001	<0.001
Male sex, no. (%)	2595 (43.5)	225 (57.3)	482 (62.9)	<0.001	<0.001
BMI, kg m ⁻²	25.52 (3.14)	26.08 (3.24)	26.10 (3.17)	<0.001	<0.001
Baseline BP, mmHg					
SBP	146.33 (16.64)	145.46 (17.35)	147.41 (16.66)	0.091	0.317
DBP	82.67 (10.61)	82.19 (11.44)	82.68 (10.40)	0.985	0.390
eGFR	109.95 (23.95)	106.76 (22.94)	108.43 (25.62)	0.102	0.010
Renal dysfunction, no. (%)					
eGFR <60 ml min ⁻¹ 1.73 m ⁻²	100 (1.4)	9 (2.3)	23 (2.5)	0.041	0.262
Fasting serum glucose, mmol l ⁻¹	5.77 (5.26–6.6)	5.76 (5.27–6.54)	5.82 (5.29–6.7)	0.696	0.524
Lipids profile, mmol l ⁻¹					
Total cholesterol	4.92 (1.10)	4.75 (1.06)	4.76 (1.07)	<0.001	0.004
Triglycerides	1.34 (0.95–1.93)	1.33 (0.95–1.95)	1.31 (0.97–1.90)	0.690	0.735
HDL-C	1.27 (0.31)	1.23 (0.30)	1.22 (0.29)	<0.001	0.008
Smoking history, no. (%)	922 (15.5)	57 (14.5)	156 (20.5)	<0.001	0.653
Medical history, no. (%)					
Diabetes mellitus	1,139 (19.1)	82 (20.9)	158 (20.6)	0.334	0.425
Hyperlipidemia	2,214 (37.1)	135 (34.4)	275 (35.9)	0.540	0.297
Cardiovascular disease	363 (6.1)	32 (8.1)	59 (7.7)	0.097	0.126
Aspirin use, no. (%)	597 (10.0)	54 (13.7)	89 (11.6)	0.185	0.023
Statin use, no. (%)	1,157 (19.4)	70 (17.8)	158 (20.6)	0.446	0.482
ARB use, no. (%)	2,620 (43.9)	175 (44.5)	358 (46.7)	0.150	0.853
The 10 year Framingham risk ≥15%, no. (%)	4,565 (77.0)	320 (81.4)	649 (85.7)	<0.001	0.047
Intensive treatment arm, no. (%)	2,988 (50.1)	205 (52.2)	377 (49.2)	0.679	0.456

Continuous variables are represented as mean ± s.d. or median (IQR), and qualitative parameters are presented as number (no., %). Continuous variables were compared using the Student's *t*-test if normally distributed or using the Wilcoxon rank-sum test if nonnormally distributed. Categorical variables were compared using the chi-square test. Two-sided *P* < 0.05 was defined as statistically significant; multiple comparisons were not adjusted. DBP, diastolic blood pressure.

Results

Descriptive characteristics

Among the 7,125 participants with available data, the median age was 65 years, 3,823 (53.66%) were female, 766 (10.75%) had preexisting CSD and 393 (5.51%) developed new-onset CSD. A total of 6,359 participants were included in the analysis of the association between treatment strategies and new-onset CSD: 3,166 in the standard treatment group and 3,193 in the intensive treatment group (Extended Data Fig. 1). Baseline characteristics were balanced between the two groups (Supplementary Table 1). As presented in Table 1, in comparison with those free of CSD, participants with new-onset CSD were older (median age 66 versus 65 years, *P* < 0.001) and more likely to be male (57.3% versus 43.5%, *P* < 0.001). Patients with new-onset CSD had higher cardiovascular risk (Framingham risk score ≥15% in 81.4% versus 77.0%, *P* = 0.047). Participants who were free of CSD had higher total cholesterol (4.92 versus 4.75 mmol l⁻¹, *P* = 0.004), high-density lipoprotein cholesterol (HDL-C) (1.27 versus 1.23 mmol l⁻¹, *P* = 0.008), but lower body mass index (BMI) (25.52 versus 26.08 kg m⁻², *P* = 0.001) compared with those with new-onset CSD.

Association of intensive BP control with new-onset CSD

Among participants without CSD at baseline (*n* = 6,359), 393 new cases of CSD occurred (205 in the intensive BP control arm and 188 in the standard BP control arm (Supplementary Table 2)) during a median (interquartile range (IQR)) follow-up of 38.5 (35.7–40.7) months. There was a similar risk of new-onset CSD in the standard and intensive

treatment arms (Table 2). Random assignment of participants to the intensive BP control arm exhibited a similar risk of first-degree atrioventricular block, LVCD and RBBB both before and after multivariable adjustment, compared with the standard BP control arm (Table 2). These results remained consistent across all sensitivity analyses (Supplementary Tables 5 and 6). In addition, the incidence risk for new pacemaker implantations was also comparable between the two treatment arms (Supplementary Table 5). In subgroup analysis, no significant interactions by age (Supplementary Table 3) or diabetes mellitus (Supplementary Table 4) were observed between treatment arms and the risk of incident CSD, first-degree atrioventricular block, LVCD and RBBB.

Association of CSD with adverse cardiovascular events

Compared with those free of CSD, participants who had preexisting CSD had a similar risk for adverse cardiovascular events (hazard ratio (HR) 1.26, 95% confidence interval (CI) 0.89–1.79 and *P* = 0.197) after adjustment for covariates. For secondary outcomes, participants who had preexisting CSD also demonstrated a similar hazard for stroke (HR 1.27, 95% CI 0.71–2.25 and *P* = 0.418), acute coronary syndrome (HR 1.34, 95% CI 0.79–2.28 and *P* = 0.279), coronary revascularization (HR 1.88, 95% CI 0.92–3.87 and *P* = 0.084), atrial fibrillation (HR 1.76, 95% CI 0.71–4.35 and *P* = 0.222), death from cardiovascular causes (HR 0.42, 95% CI 0.06–3.14 and *P* = 0.395) and death from any cause (HR 0.62, 95% CI 0.22–1.74 and *P* = 0.364) after adjustment for covariates. However, participants with preexisting CSD exhibited a greater hazard

Table 2 | Risk of incident CSD, overall and by type, comparing intensive with standard treatment^a

Variable ^b	No. of events	Model ^c	HR (95% CI)	P value
CSD	393	Unadjusted	1.08 (0.88–1.31)	0.458
		Adjusted	1.07 (0.88–1.31)	0.504
First-degree atrioventricular block	237	Unadjusted	1.04 (0.81–1.35)	0.736
		Adjusted	1.06 (0.82–1.37)	0.659
LVCD	67	Unadjusted	1.15 (0.71–1.86)	0.565
		Adjusted	1.08 (0.67–1.75)	0.741
LBBB	9	Unadjusted	1.23 (0.33–4.60)	0.755
		Adjusted	–	–
Left anterior branch block	41	Unadjusted	1.05 (0.57–1.93)	0.886
		Adjusted	0.97 (0.51–1.83)	0.915
Left posterior branch block	3	Unadjusted	–	–
		Adjusted	–	–
Nonspecific intraventricular conduction delay	14	Unadjusted	2.46 (0.77–7.87)	0.128
		Adjusted	2.82 (0.84–9.48)	0.094
RBBB	109	Unadjusted	1.13 (0.78–1.65)	0.525
		Adjusted	1.11 (0.75–1.62)	0.607

^aThe standard treatment group was the reference group and the Fine–Gray subdistribution hazard model was used for the competing risk of death. Two-sided $P < 0.05$ was defined as statistically significant; multiple comparisons were not adjusted. ^bRisk of left posterior branch block and adjusted results for LBBB are not shown due to insufficient number of events.

^cAdjusted model included age, sex, BMI, baseline SBP, smoking history, history of diabetes mellitus, history of hyperlipidemia, history of cardiovascular disease, aspirin use, statin use, ARB use, eGFR, total cholesterol and HDL-C.

for developing acute decompensated heart failure (HR 10.23, 95% CI 1.66–63.18 and $P = 0.012$) after adjustment for covariates (Table 3). Participants with new-onset CSD had a similar hazard for adverse cardiovascular events, adverse cardiovascular events separately and death from any cause compared with those free of CSD (Supplementary Table 7).

Association of clinical and demographic factors with new-onset CSD

Clinical and demographic correlates of incident CSD in the study population are shown in Fig. 1. Increased age (HR 1.29, 95% CI 1.03–1.61 and $P = 0.024$), male sex (HR 1.74, 95% CI 1.39–2.18 and $P < 0.001$) and increased BMI (HR 1.26, 95% CI 1.07–1.49 and $P = 0.006$) were positively associated with incident CSD, while increased estimated glomerular filtration rate (eGFR) (HR 0.94, 95% CI 0.90–0.98 and $P = 0.007$) and increased total cholesterol (HR 0.88, 95% CI 0.78–0.99 and $P = 0.031$) were negatively associated with incident CSD in the study. Baseline SBP, presence of diabetes mellitus, presence of hyperlipidemia, presence of cardiovascular diseases, aspirin use, statin use, angiotensin receptor blocker (ARB) use and intensive treatment were not observed to be associated with the incidence of CSD.

Discussion

In older adults with hypertension participating in the STEP trial, our findings indicate a comparable incidence of new-onset CSD between individuals randomized to intensive versus standard BP control. Additionally, our observations reveal that participants with preexisting CSD did not exhibit an increased hazard of developing adverse cardiovascular events. However, this subgroup had an elevated susceptibility to acute decompensated heart failure. New-onset CSD was predominantly associated with factors such as advanced age, male sex and higher BMI, while higher eGFR and total cholesterol levels were negatively associated with new-onset CSD.

In systemic hypertension, left ventricular pressure overload may accelerate interstitial fibrosis in the conduction system¹⁷. In addition, many signal pathways that may lead to fibrosis, including neurohormonal, cytokine, oxidative stress and inflammatory, can be activated by systemic high BP¹⁷. Most prior studies have established the association of hypertension with the incidence of CSD^{18–21}, with some post hoc analyses of randomized controlled trials observing the role of an angiotensin-converting enzyme inhibitor and ARB in decreasing the clinical incidence of CSD^{22,23}. Yet it has not been clear whether intensive BP control will further decrease the incidence of CSD. The present study shows the lack of incremental benefit of intensive BP control to reduce new-onset CSD in this older cohort with hypertension. In terms of the mechanism, interstitial fibrosis has been proposed as the driver of conduction system abnormalities, but it remains unclear why some individuals develop clinically apparent CSD while others do not²². Meanwhile, there is no well-defined CSD treatment except for pacemaker implantation.

Recent data from the SPRINT study suggested that intensive BP control reduced the incidence of LVCD but not right ventricular conduction disease in patients without diabetes mellitus²⁴. However, we observed similar incidences of LVCD and right ventricular conduction disease in both standard and intensive BP control arms among the STEP cohort with or without diabetes mellitus. One difference between these trials was the incidence of new-onset CSD^{2,25,26}. The incidence of LVCD (with 77 cases, including 9 LBBB, 41 left anterior branch block, 3 left posterior branch block and 14 intraventricular conduction delay) and RBBB (109 cases) in STEP were 1.21% and 1.71%, respectively. In the SPRINT trial, the incidences were 2.58% for LVCD (with 203 cases, including 57 LBBB, 64 left anterior branch block, 1 left posterior branch block and 81 intraventricular conduction delay) and 2.34% for RBBB (184 cases). In particular, the proportions of the various types of LVCD varied greatly between the STEP and the SPRINT cohorts. As a post hoc analysis in the STEP study, it is difficult to replicate the observed association of the new-onset CSD with the intensive BP treatment in the SPRINT study. Beyond disease incidence, several other differences between SPRINT and STEP may contribute to these discrepant results, including different age ranges, composition by race and/or ethnicity and comorbidity burden, particularly for chronic kidney disease. Future investigations are warranted to understand how these factors may impact the relationship between CSD and BP control. Furthermore, while higher BP may lead to a higher risk of CSD, stringent BP control might not completely attenuate the competing risk of CSD attributed to other uncontrolled residual factors, given its association with a range of comorbidities.

Except for the intervention difference, we hypothesized that the characteristics of the participants of the trial led to the negative result in some way. The STEP trial recruited relatively older participants with a higher baseline cardiovascular disease risk. This demographic profile suggests that the population was predisposed to CSD regardless of BP control due to continuing underlying clinical and subclinical pathology. A younger, lower-risk population with a lower comorbidity burden may be ideal for demonstrating that an intensive BP control strategy has a more notable preventive effect in reducing CSD incidence. Moreover, patients with diabetes mellitus are associated with an increased risk of CSD^{27,28}. In the STEP cohort, however, we observed that the presence of diabetes mellitus did not appear to be significantly correlated with the incidence of CSD. Potential explanations that could account for the negative findings are that we may not have adequate power to detect a statistically significant relationship, and the STEP cohort did not include patients with uncontrolled diabetes mellitus. Furthermore, glucose control may lower the risk of CSD in patients with diabetes mellitus²⁶. Diabetes mellitus may have an important influence on the relationship between intensive BP control and new-onset CSD. This hypothesis warrants further investigation, preferably in a prospective study.

Table 3 | Adverse cardiovascular events and secondary outcomes comparing participants with preexisting CSD at baseline versus those free of CSD^a

Variable	No. of events	Model ^c	β	s.e.m.	HR (95% CI)	χ^2	P value
Adverse cardiovascular events ^b	252	Unadjusted	0.427	0.174	1.53 (1.09–2.16)	2.451	0.014
		Adjusted	0.230	0.184	1.26 (0.89–1.79)	1.291	0.197
Components of adverse cardiovascular events							
Stroke	88	Unadjusted	0.453	0.291	1.57 (0.89–2.78)	1.555	0.120
		Adjusted	0.237	0.309	1.27 (0.71–2.25)	0.809	0.418
Acute coronary syndrome	108	Unadjusted	0.431	0.264	1.54 (0.92–2.58)	1.632	0.103
		Adjusted	0.293	0.277	1.34 (0.79–2.28)	1.082	0.279
Acute decompensated heart failure	6	Unadjusted	2.115	0.816	8.29 (1.67–41.05)	2.590	0.010
		Adjusted	2.325	0.965	10.23 (1.66–63.18)	2.502	0.012
Coronary revascularization	45	Unadjusted	0.731	0.373	2.08 (1.00–4.32)	1.729	0.0498
		Adjusted	0.634	0.387	1.88 (0.92–3.87)	1.729	0.084
Atrial fibrillation	35	Unadjusted	0.903	0.403	2.47 (1.12–5.43)	2.243	0.025
		Adjusted	0.564	0.443	1.76 (0.71–4.35)	1.222	0.222
Death from cardiovascular causes	17	Unadjusted	−0.655	1.031	0.52 (0.07–3.91)	−0.636	0.523
		Adjusted	−0.876	1.045	0.42 (0.06–3.14)	−0.850	0.395
Death from any cause	53	Unadjusted	−0.372	0.520	0.69 (0.25–1.91)	−0.716	0.474
		Adjusted	−0.477	0.526	0.62 (0.22–1.74)	−0.907	0.364

^aIn the analyses of the adverse cardiovascular events and secondary outcomes except for death from any cause, the HRs, 95% CIs and P values were calculated using of the Fine–Gray subdistribution hazard model for the competing risk of death. A Cox regression model was used for death from any cause. Two-sided $P < 0.05$ was defined as statistically significant; multiple comparisons were not adjusted. ^bThe adverse cardiovascular events were a composite of stroke, acute coronary syndrome, acute decompensated heart failure, coronary revascularization, atrial fibrillation or death from cardiovascular causes. ^cAdjusted models included age, sex, BMI, baseline SBP, smoking history, history of diabetes mellitus, history of hyperlipidemia, history of cardiovascular disease, aspirin use, statin use, ARB use, eGFR, total cholesterol, HDL-C and treatment strategy.

involving a population with diabetes mellitus. In addition, our study cohort consisted of older (age 60–80 years) adults with hypertension, an optimal high-risk population for further investigation of risk factors for CSD.

We found that several clinical and demographic variables beyond age are associated with the risk of future CSD. Risk factors, including increased BMI and decreased eGFR, are preventable or treatable^{22,24,28}. Further investigation is warranted to gain deeper insights into managing renal dysfunction and body weight on the incidence of CSD. Notably, our study has unveiled an intriguing observation: higher total cholesterol was associated with a reduced risk of CSD. This observation aligns with previous research findings²⁹. This unexpected result might be attributed to nearly one-fifth of those in the STEP cohort who were on statin therapy and might already have been actively modifying their lifestyles or employing lipid-lowering medications. To our surprise, smoking history was associated with a lower risk of new-onset CSD in our study. However, previous studies have reported varying results^{22,24,28}. In addition, the smoking history assessment relied on self-reporting, and some participants might not accurately disclose their smoking history. Additional research is needed to elucidate the relationship between smoking and CSD. Moreover, previous studies have found that ARB or angiotensin-converting enzyme inhibitor may mitigate the risk of CSD^{22,23}. However, our results suggested that ARB has limited effectiveness in reducing CSD risk. The status of ARB use varied within the STEP study, potentially influenced by the duration of ARB use and the efforts to control BP within target ranges. This variability may explain the absence of an association between baseline ARB use and incident CSD.

CSD and high BP independently predict adverse cardiovascular events, such as myocardial infarction, heart failure and cardiovascular death^{1,30,31}. Previous studies have focused on the association between CSD and clinical outcomes. Diverse studies have found association between various conduction abnormalities, including

LBBB^{6–8}, RBBB^{6,7,20,32}, left anterior branch block^{3,10}, left posterior branch block^{3,11}, nonspecific intraventricular conduction delay^{3,12} and first-degree atrioventricular block^{13,14}, with worsened outcomes in long-term follow-up. The precise underlying mechanisms behind these associations are uncertain. Whether conduction abnormalities are primary causes or secondary consequences of other pathologic processes, such as infarctions or fibrosis²², still requires investigation. For instance, lisinopril therapy, when compared with chlorthalidone, substantially reduced the incidence of CSD²², but lisinopril therapy was associated with an increased risk for cardiovascular disease and stroke in the Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial³³. In the STEP cohort, we have found no evidence to suggest that the presence of CSD increases the risk of adverse cardiovascular events. This can be explained by the short follow-up period. Similarly, Alventosa-Zaidin et al. reported that the presence of complete RBBB tended to increase all-cause mortality and cardiovascular events without a statistically significant difference in the relatively short follow-up³⁴. In our study, we found that the HR of cardiovascular and all-cause mortality in patients with preexisting CSD was less than 1. There may probably be a survival effect when we analyze the relationship between preexisting CSD and death. That is, if a participant had died before completing a follow-up ECG, the participant would have been excluded from our current analysis due to a lack of follow-up ECG. Participants with preexisting CSD may have a lack of follow-up ECG due to fatal events; therefore, we may have underestimated mortality in patients with preexisting CSD in the current analysis. However, we observed that participants with preexisting CSD had a greater risk of acute decompensated heart failure in the STEP cohort. The competing risk from coexisting comorbidities along with hypertension may play an essential role in this complex relationship.

Hence, further research is essential to elucidate the role of intensive BP control in preventing CSD. This necessitates studies with extended follow-up periods involving younger participants,

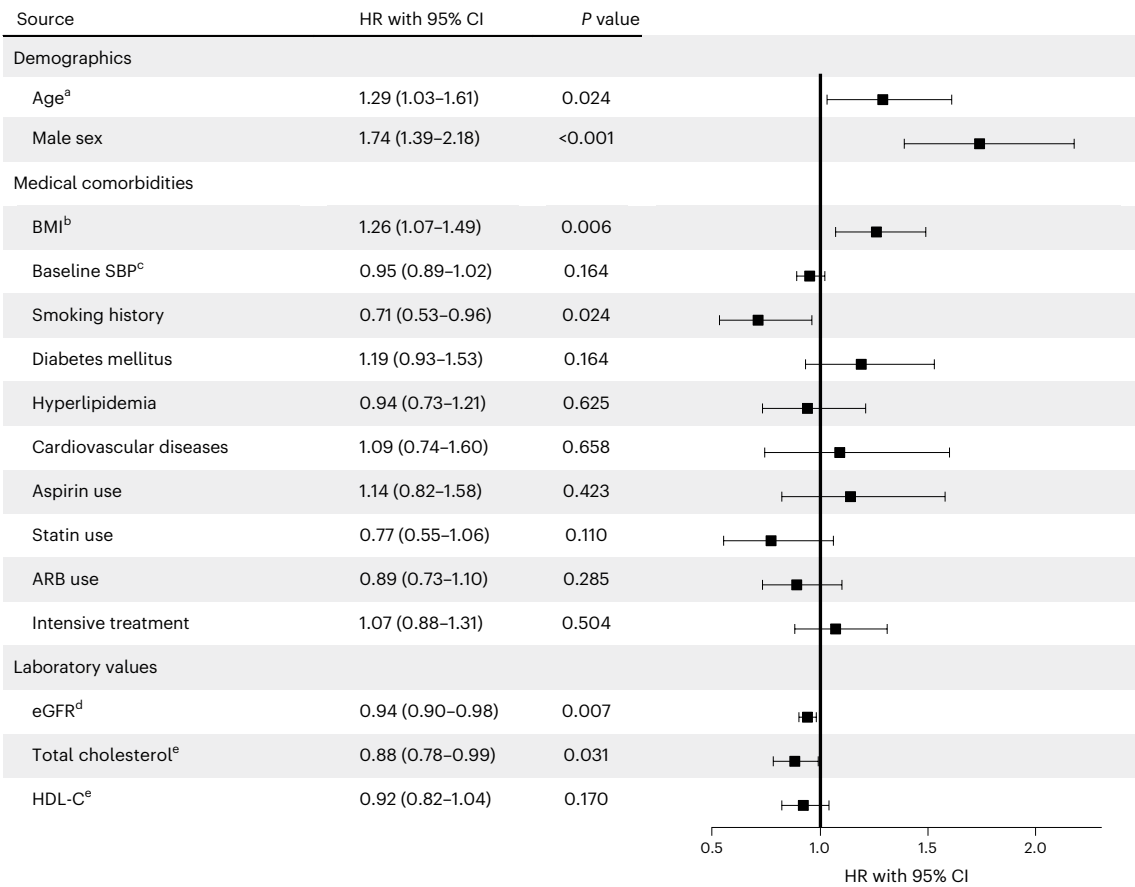


Fig. 1 | Multivariable clinical predictors of incident CSD. A multivariable-adjusted Fine–Gray subdistribution hazard model was used to evaluate the hazard of new-onset CSD. Two-sided $P < 0.05$ was defined as statistically significant; multiple comparisons were not adjusted. Adjusted models included all covariates listed. Data are presented as point estimates of effects ± 1.96 s.e.m.

(that is, 95% CI). The nodes represent the point estimates, the line segment represents the CI and the solid line is the reference line. Analyses were performed based on the data of participants without CSD at baseline ($n = 6,359$). ^aIndicates per 10 year increase. ^bIndicates per 5 kg m⁻² increase. ^cIndicates per 10 mmHg increase. ^dIndicates per 10 ml min⁻¹ 1.73 m⁻² increase. ^eIndicates per s.d. increase.

particularly those with diabetes mellitus or overweight, to determine the long-term effects of intensive BP control in reducing the CSD burden among hypertensive individuals. Identifying optimal BP treatment goals is crucial for reducing the overall CSD burden in this population. Therefore, additional research endeavors are warranted to gain comprehensive insights into the role of intensive BP control in the secondary prevention of CSD.

Study limitations

There are several limitations in our study. First, our results are constrained by the relatively short follow-up period and the less frequent ECG recordings to detect incident CSD. Second, our study population consisted of older Chinese individuals aged 60–80 years, characterized by a high comorbidity burden. This demographic profile could have affected our ability to observe a substantial effect of BP control on CSD and decreased the generalizability of our results to other cohorts. Third, brain natriuretic peptide, a marker of left ventricular wall stress, which may be crucial in the occurrence of CSD, was not measured in the STEP study. Additionally, there may have been a survival effect when analyzing the association of CSD with cardiovascular mortality and all-cause mortality, potentially introducing bias into the results. Last, CSD was not a primary end point of the STEP trial, and our study was a post hoc analysis. Despite these limitations, the robust study methodology employed by the STEP trial investigators has provided valuable insights into the role of intensive hypertension management in the incidence and impact of CSD.

Conclusions

In older adults with hypertension, intensive BP control may not reduce the incidence of new-onset CSD compared with standard BP control. Participants with preexisting CSD face a greater risk for acute decompensated heart failure regardless of intensive BP control.

Methods

This post hoc analysis of a randomized clinical trial followed the Strengthening the Reporting of Observational Studies in Epidemiology reporting guidelines. Data collection and evaluation were approved by the ethics committees of FuWai Hospital and collaborating centers, and all the participants provided written informed consent.

Study population and design

The STEP trial (ClinicalTrials.gov number [NCT03015311](https://clinicaltrials.gov/ct2/show/study?term=NCT03015311)) was a prospective, randomized, controlled trial that was conducted at 42 clinical centers throughout China. The design, process and rationale of the STEP trial have been published previously^{35,36}. Briefly, the STEP trial aimed to test whether reducing systolic blood pressure (SBP) to 110–130 mmHg reduces adverse cardiovascular events, defined as a composite of stroke (ischemic or hemorrhagic), acute coronary syndrome (acute myocardial infarction and hospitalization for unstable angina), acute decompensated heart failure, coronary revascularization, atrial fibrillation or death from cardiovascular causes. Patients were included if they were aged 60–80 years and were hypertensive with SBP of 140–190 mmHg at the screening visit. Participants with a history of

ischemic or hemorrhagic stroke, mental impairment, uncontrolled diabetes mellitus or a severe life-limiting condition were excluded. Details about the inclusion and exclusion criteria have been published elsewhere³⁵.

A total of 8,511 participants were enrolled from 10 January to 31 December 2017, of whom 4,243 were randomized to the SBP target of 110–130 mmHg (intensive treatment arm) and 4,268 participants were randomized to the SBP target of 130–150 mmHg (standard treatment arm). In the STEP trial, patients were provided with antihypertensive drugs, including olmesartan (an ARB), amlodipine (a calcium channel blocker) and hydrochlorothiazide (a diuretic) for free. Patients commenced treatment with olmesartan as the preferred ARB or amlodipine besylate tablets as the preferred calcium channel blocker. Hydrochlorothiazide was not used as the initial therapy. Other antihypertensive drugs were permitted if deemed necessary and appropriate by the investigator. The STEP trial intervention was stopped early because of an apparent reduction in the primary composite cardiovascular disease in the intensive treatment arm at two consecutive time points.

Among the 8,511 participants, 8,064 had a baseline ECG recorded (in 2017), 6,788 underwent ECG assessments at year 2 (2019–2020) and 6,322 had an ECG at year 3 or study closeout (2020–2021). The follow-up ECG was defined as the last available one for each patient. For example, if a participant had a follow-up ECG at year 2 and then again at year 3 or study closeout, the ECG at year 3 or study closeout was selected for our analysis. Finally, 7,125 participants, including 3,823 females, were included in the final analysis (Extended Data Fig. 1).

Ascertainment of CSD

CSD, including first-degree atrioventricular block, LBBB, RBBB, left anterior branch block, left posterior branch block and nonspecific intraventricular conduction delay, were defined by the 2018 American College of Cardiology/American Heart Association/Heart Rhythm Society guidelines⁴⁵. We ascertained CSD from standard 12-lead ECG tracings obtained at baseline (2017), year 2 (2019–2020) and year 3 or study closeout (2020–2021). Digital ECG data were recorded at 10 mm mV⁻¹ calibration and a speed of 25 mm s⁻¹ at FuWai Hospital and collaborating centers. ECG readings were performed by trained staff who were blinded to treatment allocation at the Shanghai Tenth People's Hospital.

Covariates

The sociodemographic characteristics of participants were collected at baseline, including age, sex, BMI and smoking history. The 10 year risk of cardiovascular disease was calculated using the Framingham risk score³⁷. Those who had smoked at least one cigarette per day for more than 6 months were considered to smoking history. BMI was calculated using the formula: weight (kg)/height squared (m²).

Clinical information, including the history of diabetes mellitus, hyperlipidemia, cardiovascular disease and drug history, was collected at baseline. Office BP was measured using the same validated automatic BP monitor device (Omron HBP-1100U, Omron Healthcare) at baseline and each visit. Before BP was measured, the participants were required to rest for at least 5 min. The BP was measured three times with 1–2 min intervals, and the average value was recorded by a qualified nurse or physician. Laboratory examinations, such as fasting serum glucose, eGFR, triglycerides, total cholesterol and HDL-C, were conducted at baseline.

Outcomes

The primary outcome for this post hoc analysis was the incidence of new-onset CSD in those free of preexisting CSD, stratified by treatment strategy. Secondary outcomes included the individual components of the CSD, including first-degree atrioventricular block, LVCD (composite of LBBB, left anterior branch block, left posterior branch block and nonspecific intraventricular conduction delay) and RBBB.

The additional study outcome for this post hoc analysis was the risk of development of the adverse cardiovascular events (composite of stroke, acute coronary syndrome, acute decompensated heart failure, coronary revascularization, atrial fibrillation or death from cardiovascular causes), adverse cardiovascular events separately and death from any cause in participants with preexisting CSD or new-onset CSD.

Statistics and reproducibility

A central computerized randomization program on a Web-based interface was used to randomize with stratification according to clinical sites. Investigators and patients were aware of the trial group assignments, but all outcomes were evaluated by an adjudication committee blinded to treatment allocation. We excluded participants with insufficient ECG data, including those without qualified ECG images at baseline and follow-up, and those with pacemaker or ventricular preexcitation detected on baseline ECG. No statistical methods were used to predetermine sample sizes but our sample sizes are similar to those reported in previous publications^{23,24}.

Baseline characteristics were compared by CSD status using descriptive statistics. Participants without preexisting CSD were compared and grouped by treatment arms using descriptive statistics. Continuous variables were summarized as mean $\pm$ s.d. and compared using the Student's *t*-test if normally distributed, or median (IQR) and compared using the Wilcoxon rank-sum test if nonnormally distributed. Categorical variables were summarized as absolute numbers (percentage) and compared using the chi-square test.

Univariable and multivariable-adjusted Fine–Gray subdistribution hazard models stratified by clinical sites were used to account for the competing risk of death and assess the association between treatment arms and incident CSD³⁸. The multivariable model was adjusted for age, sex, BMI, baseline SBP, smoking habit, presence of diabetes mellitus, presence of hyperlipidemia, presence of cardiovascular diseases, aspirin use, statin use, ARB use, eGFR, total cholesterol, HDL-C and treatment strategy. Mean imputation was used for missing data of covariates. Several sensitivity analyses were performed. First, incident pacemakers detected by follow-up ECG scans were included in the outcome. Second, inverse probability weighting was used to account for missing data of follow-up ECG³⁹. Subgroup analyses on the associations of treatment arms with CSD were conducted with stratified factors, including age (<mean age versus $\geq$ mean age years) and diabetes mellitus (yes versus no). We also performed separate analyses for every category of CSD, including LVCD, RBBB and first-degree atrioventricular block.

The Fine–Gray subdistribution hazard model with stratification of clinical sites was used to account for the competing risk of death and evaluate the hazard of the adverse cardiovascular events and secondary outcomes except for death from any cause, in participants with preexisting or new-onset CSD³⁸. The new-onset CSD was cast as a time-varying covariate. The Cox regression model with stratification of clinical sites was used for analyzing death from any cause. Model 1 was unadjusted, while model 2 was adjusted for age, sex, BMI, baseline SBP, smoking habit, presence of diabetes mellitus, presence of hyperlipidemia, presence of cardiovascular diseases, aspirin use, statin use, ARB use, eGFR, total cholesterol, HDL-C and treatment strategy.

A multivariable-adjusted Fine–Gray subdistribution model with stratification of clinical sites was used to evaluate the association between the risk of new-onset CSD and various clinical and demographic factors. The model was adjusted for the above-mentioned covariates.

All statistical analyses were performed using R software, version 4.1.2 (R Foundation for Statistical Computing), with a two-tailed *P* value of <0.05 considered statistically significant.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data are not publicly available due to restrictions of ethical approval requirements and the current national general data protection regulations of this study. Instructions for applying for data access are detailed at https://www.nejm.org/doi/suppl/10.1056/NEJMoa2111437/suppl_file/nejmoa2111437_data-sharing.pdf. The complete de-identified patient data set underlying this article will be shared on reasonable request to the corresponding author.

Code availability

Survival 3.3.1 in R version 4.1.2 was used to perform Cox regression model and Fine–Gray subdistribution hazard model, and to conduct inverse probability weighting. Complete codes of R software (version 4.1.2) are available from the corresponding author on reasonable request.

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Author contributions

Y.Z. and J.C. designed the study. S.Z. drafted the paper. Y.D. and Y.W. coded and analyzed the data. Y.Z. and J.C. supervised the study and revised the paper. S.Y. and J.H. revised the paper for important intellectual content.

Competing interests

The authors declare no competing interests.

Additional information

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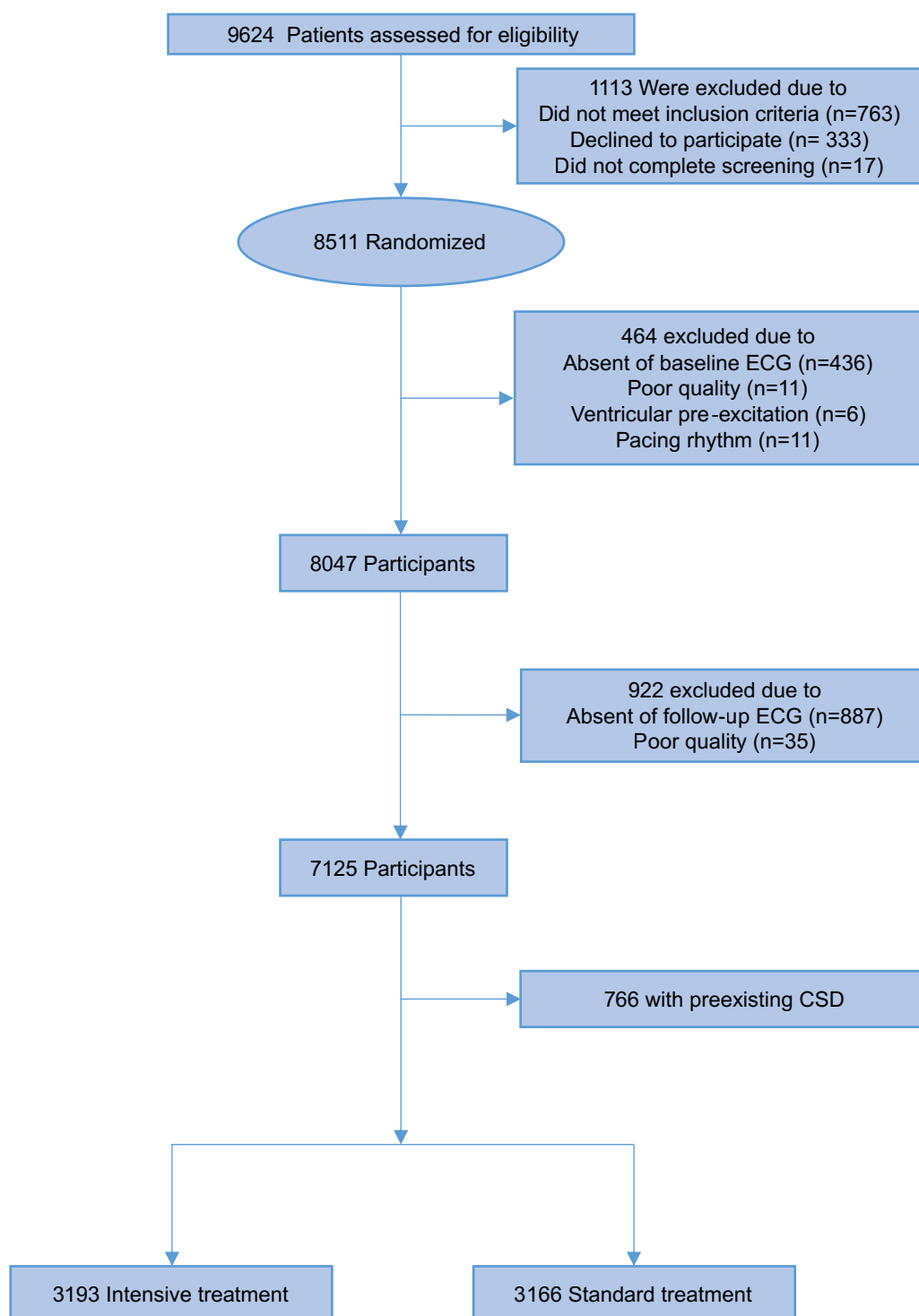
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Extended Data Fig. 1 | The flow of the study sample. Abbreviations: ECG, electrocardiogram; CSD, conduction system diseases.

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☐	☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
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Data collection	No software was used.
Data analysis	R version 4.1.2 package: Survival 3.3.1 was used to perform Cox regression model and Fine–Gray subdistribution hazard model, and to conduct inverse probability weighting. Complete codes of R software (version 4.1.2) are available from the corresponding author on reasonable request.

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Reporting on sex and gender	We used the sex term.
Reporting on race, ethnicity, or other socially relevant groupings	All participants were Han ethnicity in the study.
Population characteristics	Among 7125 participants with available data, the median age was 65 years, 3823 (53.66%) were female, 766 (10.75%) had preexisting conduction system diseases, and 393 (5.51%) developed new-onset conduction system diseases.
Recruitment	A total of 8511 participants were enrolled from January 10 to December 31, 2017, of whom 4243 were randomized to the SBP target of 110 to <130 mm Hg (intensive-treatment arm), and 4268 participants were randomized to the SBP target of 130 to <150 mm Hg (standard-treatment arm).
Ethics oversight	Data collection and evaluation were approved by the ethics committees of FuWai Hospital and collaborating centers.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

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Sample size	Among 8511 participants, 8064 participants had baseline electrocardiogram recorded (2017), 6788 participants underwent electrocardiogram assessments at year 2 (2019-2020) and 6322 participants had electrocardiogram at year 3 or study closeout (2020-2021). The follow-up electrocardiogram was defined as the last available one for each patient. For example, if a participant had a follow-up ECG at year 2 and then again at year 3/closeout, the electrocardiogram at year 3 or closeout was selected for our analysis. Finally, 7125 participants were included in the final analysis. This is a post hoc analysis of a randomized clinical trial. No statistical methods were used to pre-determine sample sizes but our sample sizes are similar to those reported in previous publications (J Hypertens 37, 629-635 (2019); JAMA Cardiol 8, 612-616 (2023)).
Data exclusions	We excluded participants with insufficient electrocardiogram data, including those without qualified ECG images at baseline and follow-up, and those with pacemaker or ventricular pre-excitation detected on baseline electrocardiogram .
Replication	This is a post hoc analysis of a randomized clinical trial, and only one baseline and one follow-up electrocardiogram were selected for analysis. Therefore, We did not repeat the analysis.
Randomization	At each center, eligible patients were randomly assigned in a 1:1 ratio to a systolic blood-pressure target of 110 to less than 130 mmHg (intensive treatment) or a target of 130 to less than 150 mmHg (standard treatment).Randomization was performed with the use of a central computerized randomization program on a Web-based interface, with stratification according to clinical center.
Blinding	Investigators and patients were aware of the trial group assignments, but all outcomes were evaluated by an adjudication committee blinded to treatment allocation. ECG reading was performed by trained staff who blinded to treatment allocation at the Shanghai Tenth People's Hospital.

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